

EE6203 Quiz 1 — Evaluation Set

Worked Solutions and Score Interpretation

Companion to `quizzes/quiz-01-evaluation-set.html`

31 August 2026

Before you open this

This document contains the answers. It is separate from the question file for a reason: the evaluation set is a *measurement*, and a measurement taken with the answer key in view measures nothing. Sit all 44 questions closed-book first, with a calculator and Table 2–1 available, then come here.

Scope is Chapters 1–3 only, matching the Week 3 announcement that Quiz 1 is a 30-minute closed-book assessment excluding the Chapter 4 material. Every item is new; none is reused from the drills, knowledge checks, worked archetypes or mock quizzes in `notes/quiz1-prep-learning` or `notes/quiz1-prep-reviewing`.

Each numeric answer below was derived twice by independent routes — a transform argument and a direct time-domain or physical argument — and the two agree. Where the second route is quick enough to use under exam pressure, it is given as a **Check**, because that is the habit worth carrying into the quiz.

How to read your topic scores

The per-topic bars matter more than the total. Treat a topic below 60 % as a gap, 60–79 % as shaky, and 80 %+ as sound. Then fix gaps in dependency order, not in alphabetical order — later topics are built out of earlier ones, and repairing **H** while **C** is broken wastes the effort.

Tier	Topics	Why it comes first
1	B, C	The transform definition and the shift/translation theorems. Everything downstream is algebra on top of these. A gap here shows up as wrong answers in D, E, F and H that look like different mistakes but are not.
2	D, E, F	Reading a transform back: legality of the final value, inverse methods, recurrences. These are the most heavily examined single-topic items.
3	G, I	Impulse sampling, the starred transform, and the star-factorisation rule $[GX^*]^* = G^*X^*$. This one rule governs every block-diagram question.
4	H, J	ZOH-plus-plant conversion and the closed-loop configurations. Highest mark density in past questions, but only reachable once tiers 1–3 hold.
—	A	Vocabulary. Cheap marks; revise by reading, not by drilling.

Minimum viable path. If time is short, the shortest route to a safe pass is C2 (advance with initial conditions) → D3/D4 (final-value legality) → E3 (repeated-pole inversion) → H4 and H1 (the $(1-z^{-1})\mathcal{Z}\{G(s)/s\}$ conversion) → G2 (star factorisation) → J1/J2 (which denominator). Those seven skills, done reliably, cover the structure of every recalled past question. Everything else in this set is either supporting arithmetic or vocabulary.

What you may safely defer. A wrong answer on A2, B3 or G3 costs almost nothing to repair and does not block anything else. Deferring E1 (method choice) is also acceptable if D and E are otherwise strong, since the wrong method still reaches the right answer, just slower.

Answer summary

Mark yourself against this table first, then read only the items you missed. Multiple-choice answers are quoted by content, because the question file shuffles the option order.

Item	Topic	Answer
A1	A	Digital (discrete in time <i>and</i> amplitude)
A2	A	The discrete and continuous descriptions must be reconciled in one domain
A3	A	26 sample instants
B1	B	$2 + 2z^{-1} + 2z^{-2} - z^{-3}$
B2	B	$z = e^{-1} \approx 0.3679$
B3	B	$T^2 z^{-1}(1 + z^{-1})/(1 - z^{-1})^3$
B4	B	$x[4] = 0.5$
C1	C	$z^{-2}/(1 - 0.4z^{-1})$
C2	C	$z^2 X(z) - 3z^2 + z$
C3	C	Every z^{-1} scaled by e^{-aT} , so z^{-2} carries e^{-2aT}
C4	C	$z = 1.2$
D1	D	$x[0] = 0$
D2	D	$x[1] = 2$
D3	D	Theorem applies; $x[\infty] = 0.5/1.9 = 0.263$
D4	D	Theorem does not apply; the remaining pole 1.2 is outside, so $x[k]$ diverges
D5	D	Two poles at $z = 1$ (double), so the theorem fails
E1	E	Direct division (long division)
E2	E	$x[2] = 0.4375$
E3	E	$x[k] = k(0.8)^{k-1}$
E4	E	$x[3] = 1.92$
E5	E	$x[k] = \sin(0.5k)$
F1	F	$y[3] = 2.56$
F2	F	$Y(z) = z/[(z - 0.5)(z - 0.6)]$
F3	F	$y[2] = 1.1$
F4	F	$y[k] = 10[(0.6)^k - (0.5)^k]$
G1	G	Impulse strength $e^{-1} \approx 0.3679$
G2	G	$[G(s)X^*(s)]^* = G^*(s)X^*(s)$
G3	G	$\omega_s = 2\pi/T \approx 314.16$ rad/s
G4	G	$X(z) = 1/(1 - z^{-1})$
H1	H	$0.8647z^{-1}/(1 - 0.1353z^{-1})$
H2	H	$G(1) = 1$
H3	H	$y[1] = 1 - e^{-2} \approx 0.8647$
H4	H	$X(z) = (1 - z^{-1})\mathcal{Z}\{G(s)/s\}$
H5	H	$g[1] \approx 0.2838$
I1	I	0.1917
I2	I	$G(z)H(z) \neq \mathcal{Z}\{G(s)H(s)\}$; the intermediate sampler decides
I3	I	No — R cannot be separated from the dynamics
I4	I	Causality of $g(t)$
J1	J	$C(z) = G(z)R(z)/[1 + GH(z)]$
J2	J	$C(z) = G(z)R(z)/[1 + G(z)H(z)]$
J3	J	$G_1 R(z)$ does not factor, so no input-independent ratio exists
J4	J	$K_P = 1.8$
J5	J	$G_D(z) = K_P + K_I/(1 - z^{-1}) + K_D(1 - z^{-1})$

Item	Topic	Answer
J6	J	$m[0] = 6.2$

1 Worked solutions

1.1 A — Signals and the digital-control architecture

A1. [PDF pp. 13–16]

A signal is *sampled-data* when time is discrete but amplitude is still continuous, and *digital* when both are discrete. A 12-bit converter has 2^{12} representable levels, so the amplitude is quantised: the ADC output is digital.

If you missed it. You are treating “sampled” and “digital” as synonyms. The distinction is the one Chapter 1 spends three slides on, and it is the reason quantisation is ignored for the rest of the course — with enough bits the digital signal is *modelled* as sampled-data, but that is an approximation, not an identity.

A2. [PDF pp. 10, 16]

Figure 1.2 shows the loop crossing a boundary twice: the ADC turns the continuous error $e(t)$ into the sequence $e[k]$, and the DAC-plus-ZOH turns the computed $u[k]$ back into a continuous $u(t)$ that drives the plant. The controller is therefore discrete while the plant remains continuous, and the two descriptions cannot be multiplied together as they stand.

Reconciling them is precisely the work of the next two chapters: Chapter 2 builds the discrete description (the z transform), and Chapter 3 converts the continuous plant, together with its hold, into a pulse transfer function in the same variable. Only then does a single algebra cover the whole loop.

If you missed it. The clock synchronises the conversions; it does not abolish the boundary. And nothing here asks for the plant to be redesigned — the plant is whatever physics gives you. If this framing is new, it is worth fixing early: every topology question in I and J is an instance of it.

A3. [PDF pp. 14–16]

$0.1/0.004 = 25$ sampling intervals, hence $k = 0, 1, \dots, 25$, which is **26** sample instants.

If you missed it. Answering 25 is the same off-by-one that corrupts piecewise-input transforms: counting intervals instead of instants. On a half-open interval $0 \leq t < nT$ there are n samples; on a closed interval $0 \leq t \leq nT$ there are $n + 1$.

1.2 B — Z-transform definition and elementary functions

B1. [PDF pp. 24–25]

Tabulate before summing:

k	0	1	2	3	4	≥ 5
$x[k]$	2	2	2	-1	0	0

The interval $0 \leq t < 3T$ is half-open, so it holds $k = 0, 1, 2$; the sample at $k = 3$ already belongs to the next piece. Substituting into $X(z) = \sum_{k \geq 0} x[k]z^{-k}$,

$$X(z) = 2 + 2z^{-1} + 2z^{-2} - z^{-3}.$$

No table entry and no shift theorem is used; the coefficients *are* the samples.

B2. [PDF pp. 32, entry 4]

Entry 4 reads $e^{-akT} \leftrightarrow 1/(1 - e^{-aT}z^{-1})$. With $a = 2$ and $T = 0.5$, $aT = 1$, so the single pole is at $z = e^{-1} = 0.3679$.

Check. The sequence is $x[k] = e^{-2(0.5k)} = (e^{-1})^k$, a geometric sequence of ratio e^{-1} ; a geometric sequence of ratio r has its pole at $z = r$.

If you missed it. Answering -2 or e^{-2} confuses a with aT . The pole depends on the sampling period; the same continuous plant gives a different discrete pole at a different T .

B3. [PDF pp. 32, entry 6]

$\mathcal{Z}\{t^2\}$ sampled at period T is $T^2 z^{-1}(1 + z^{-1})/(1 - z^{-1})^3$. The $(1 + z^{-1})$ numerator factor is the part most often lost. The entry $Tz^{-1}/(1 - z^{-1})^2$ is the ramp kT .

Check. Expand the first two coefficients: $T^2 z^{-1}(1 + z^{-1})(1 + 3z^{-1} + \dots) = T^2 z^{-1} + 4T^2 z^{-2} + \dots$, matching $(kT)^2$ at $k = 1, 2$: T^2 and $4T^2$.

B4. [PDF pp. 33, entry 20]

The pair is $ka^{k-1} \leftrightarrow z^{-1}/(1 - az^{-1})^2$. With $a = 0.5$ and $k = 4$,

$$x[4] = 4(0.5)^3 = 0.5.$$

If you missed it. Reading the pair as ka^k gives $4(0.5)^4 = 0.25$. The $k - 1$ exponent is what makes the sequence start at $x[1] = 1$ rather than $x[1] = a$; checking the first sample against the transform's leading coefficient catches this instantly.

1.3 C — Properties and the translation theorems

C1. [PDF pp. 35]

The shifting theorem gives $\mathcal{Z}\{x[k - n]\mathbf{1}[k - n]\} = z^{-n}X(z)$, so with $X(z) = 1/(1 - 0.4z^{-1})$,

$$\mathcal{Z}\{x[k - 2]\mathbf{1}[k - 2]\} = \frac{z^{-2}}{1 - 0.4z^{-1}}.$$

A delay changes *where* the sequence starts; it does not move the pole.

If you missed it. The variant $0.16/(1 - 0.4z^{-1})$ applies the complex-translation theorem instead, i.e. it scales the amplitude as a^k would. Keep the two apart: z^{-n} shifts in time, $X(z/a)$ scales in amplitude.

C2. [PDF pp. 35–36]

For the one-sided transform,

$$\mathcal{Z}\{x[k + n]\} = z^n \left[X(z) - \sum_{j=0}^{n-1} x[j]z^{-j} \right].$$

With $n = 2$, $x[0] = 3$, $x[1] = -1$:

$$z^2[X(z) - 3 - (-1)z^{-1}] = z^2X(z) - 3z^2 + z.$$

If you missed it. The sign of the last term is the whole question. An advance *removes* the samples the sequence has already produced, so each removed sample enters with a minus sign — and a negative sample therefore enters with a plus. Every forced recurrence in topic F depends on getting this right.

C3. [PDF pp. 39–40; 33, entry 16]

The complex-translation theorem states $\mathcal{Z}\{e^{-akT}x[k]\} = X(e^{aT}z)$, which in terms of z^{-1} means: replace every z^{-1} by $e^{-aT}z^{-1}$. Applying that to the sine pair,

$$\frac{e^{-aT}z^{-1} \sin \omega T}{1 - 2e^{-aT}z^{-1} \cos \omega T + e^{-2aT}z^{-2}},$$

where the z^{-2} term picks up e^{-2aT} because it carries two factors of z^{-1} . This is entry 16 of Table 2–1.

If you missed it. Scaling only the numerator is a half-application. The rule is mechanical — substitute, do not patch — and the e^{-2aT} on the z^{-2} term is the marker that it was done properly.

C4. [PDF pp. 34, 39]

$\mathcal{Z}\{a^kx[k]\} = X(z/a)$. A pole of the new transform occurs where the argument equals an old pole: $z/2 = 0.6$, hence $z = 1.2$.

If you missed it. Dividing instead of multiplying gives 0.3. Note what happened physically: multiplying a sequence by 2^k pushed a stable pole outside the unit circle. That is the same mechanism that makes $|a| \geq 1$ forcing terms diverge in topic D.

1.4 D — Initial- and final-value theorems

D1–D2. [PDF pp. 42, 51–53]

The initial-value theorem is $x[0] = \lim_{z \rightarrow \infty} X(z)$. Every numerator term of

$$X(z) = \frac{2z^{-1} + 3z^{-2}}{1 - 0.5z^{-1} + 0.06z^{-2}}$$

carries a negative power of z , so the numerator vanishes as $z \rightarrow \infty$ while the denominator tends to 1. Hence $x[0] = 0$.

For $x[1]$, multiply out and match coefficients of z^{-1} :

$$x[1] - 0.5x[0] = 2 \implies x[1] = 2.$$

Check. Continue the recursion, matching each coefficient in turn:

$$x[k] = (\text{numerator})_k + 0.5x[k-1] - 0.06x[k-2] \implies 0, 2, 4, 1.88, 0.70, 0.2372, \dots$$

The sequence decays, as the poles at 0.2 and 0.3 require.

If you missed it. Reading the leading numerator coefficient as $x[0]$ gives 2, which is actually $x[1]$. The initial-value theorem is about the z^0 coefficient, and this numerator has none.

D3. [PDF pp. 44–46]

The theorem (p. 44) requires all poles of $X(z)$ inside the unit circle, *with the possible exception of a simple pole at $z = 1$* . Form

$$(1 - z^{-1})X(z) = \frac{0.5z^{-1}}{1 + 0.9z^{-1}},$$

whose only pole is $z = -0.9$. Since $|-0.9| < 1$, the theorem applies and

$$x[\infty] = \lim_{z \rightarrow 1} \frac{0.5z^{-1}}{1 + 0.9z^{-1}} = \frac{0.5}{1.9} = 0.263.$$

If you missed it. Rejecting the theorem because -0.9 is negative confuses sign with magnitude. A negative pole gives an *alternating* transient, and an alternating transient still decays provided its magnitude is below one. Only $|\cdot| = 1$ (sustained alternation, as at $z = -1$) or > 1 destroys the limit.

D4. [PDF pp. 44–46]

Removing the step pole,

$$(1 - z^{-1})X(z) = \frac{z-1}{z} \cdot \frac{z}{(z-1)(z-1.2)} = \frac{1}{z-1.2},$$

which has its pole at $z = 1.2$, outside the unit circle. The theorem therefore does not apply, and indeed $x[k]$ contains a $(1.2)^k$ mode and diverges. The value -5 obtained by substitution is arithmetically correct and physically meaningless.

If you missed it. This is the pitfall the Remarks slide (p. 46) warns about in exactly these words: the theorem “can yield misleading results” for unbounded or oscillatory sequences. The discipline is fixed — cancel the permitted pole at $z = 1$ first, classify what survives, and only then take the limit. Never take the limit first.

D5. [PDF pp. 32 entry 5; 44]

$X(z) = Tz^{-1}/(1 - z^{-1})^2$ has a *double* pole at $z = 1$. The theorem permits at most a simple one, so it does not apply — correctly, since $x[k] = kT$ grows without bound.

If you missed it. If you answered 1, you read $(1 - z^{-1})^2$ as a single factor. The exponent is the pole multiplicity, and multiplicity is exactly what separates the step (allowed) from the ramp (not allowed).

1.5 E — Inverse z-transform methods

E1. [PDF pp. 49–53]

Long division produces $x[0], x[1], x[2], \dots$ directly, one coefficient per step, and never yields a general term. Partial-fraction expansion of $X(z)/z$ yields the closed form but costs a factorisation and a residue calculation. With only four samples wanted and no closed form required, long division is faster.

If you missed it. Choosing partial fractions here is not *wrong*, only slow — and in a 30-minute closed-book quiz, slow is a mark. Match the method to what is asked for: “find $x[k]$ ” means PFE, “find the first few samples” means long division.

E2. [PDF pp. 60–62]

Write $u = z^{-1}$ and expand $1/[(1 - 0.5u)(1 - 0.25u)] = A/(1 - 0.5u) + B/(1 - 0.25u)$. Evaluating at $u = 2$ gives $A = 2$; at $u = 4$ gives $B = -1$. Hence $x[k] = 2(0.5)^k - (0.25)^k$, and

$$x[2] = 2(0.25) - 0.0625 = 0.4375.$$

Check. Long division of $1/(1 - 0.75z^{-1} + 0.125z^{-2})$ gives 1, 0.75, 0.4375, 0.234375, matching the closed form term by term. Two independent routes agreeing is the standard for anything you will write down under time pressure.

E3–E4. [PDF pp. 33 entry 20; 61]

Convert to the table's variable before matching:

$$X(z) = \frac{z}{(z - 0.8)^2} = \frac{z^{-1}}{(1 - 0.8z^{-1})^2} \longleftrightarrow x[k] = k(0.8)^{k-1}.$$

Then $x[3] = 3(0.8)^2 = 1.92$.

Check. Long division of $z^{-1}/(1 - 1.6z^{-1} + 0.64z^{-2})$ gives 0, 1, 1.6, 1.92, 2.048. Note the sequence *rises* before decaying — the k factor beats the 0.8^k decay until $k \approx 4.5$, which is what a repeated pole inside the unit circle looks like.

If you missed it. The two index-shifted variants $k(0.8)^k$ and $(k+1)(0.8)^k$ both give the wrong $x[1]$: the true value is 1, they give 0.8 and 1.6. Testing a candidate closed form at $k = 0$ and $k = 1$ costs five seconds and settles the question.

E5. [PDF pp. 33 entry 14; 63–64]

Entry 14 is $\sin \omega kT \leftrightarrow z^{-1} \sin \omega T / (1 - 2z^{-1} \cos \omega T + z^{-2})$. Comparing, $\omega T = 0.5$, so $x[k] = \sin(0.5k)$.

If you missed it. A cosine would show $1 - z^{-1} \cos \omega T$ in the numerator; a damped sine would carry e^{-aT} factors (entry 16, as in C3). Recognising the shape of the denominator $1 - 2z^{-1} \cos \omega T + z^{-2}$ — unit leading and trailing coefficients, poles on the unit circle — is faster than any expansion.

1.6 F — Difference equations solved by z transform

F1. [PDF pp. 67–69]

Transform the homogeneous equation with its initial condition:

$$zY(z) - zy[0] - 0.8Y(z) = 0 \implies Y(z) = \frac{5z}{z - 0.8}, \quad y[k] = 5(0.8)^k.$$

Hence $y[3] = 5(0.512) = 2.56$.

Check. Iterate the rule directly: 5, 4, 3.2, 2.56.

If you missed it. If a $1/(1 - z^{-1})$ factor appeared in your working, you added a forcing term that is not there. An unforced recurrence has no step pole and therefore no constant steady-state component; it decays to zero.

F2–F4. [PDF pp. 35–36, 60, 67–69]

For a two-step advance, $\mathcal{Z}\{y[k+2]\} = z^2Y(z) - z^2y[0] - zy[1]$. With $y[0] = 0$, $y[1] = 1$:

$$z^2Y - z - 1.1zY + 0.3Y = 0 \implies Y(z) = \frac{z}{z^2 - 1.1z + 0.3} = \frac{z}{(z - 0.5)(z - 0.6)}.$$

The factorisation checks out because $0.5 + 0.6 = 1.1$ and $0.5 \times 0.6 = 0.3$.

Directly from the rule, $y[2] = 1.1y[1] - 0.3y[0] = 1.1$.

For the closed form, expand $Y(z)/z$:

$$\frac{Y(z)}{z} = \frac{1}{(z - 0.5)(z - 0.6)} = \frac{-10}{z - 0.5} + \frac{10}{z - 0.6}, \quad y[k] = 10[(0.6)^k - (0.5)^k].$$

Check. Test the closed form at three points: $y[0] = 10(1-1) = 0 \checkmark$, $y[1] = 10(0.1) = 1 \checkmark$, $y[2] = 10(0.36-0.25) = 1.1 \checkmark$ — matching the initial conditions and the sample generated from the recurrence. The full opening is 0, 1, 1.1, 0.91, 0.671, 0.4651, decaying as both poles are inside the unit circle.

If you missed it. A numerator of 1 or z^2 in F2 means the initial-condition terms were mishandled — go back to C2, which is the same rule in isolation. In F4, the three-point test rejects every wrong option in under a minute, so there is no excuse for guessing on a closed form.

1.7 G — Impulse sampling and the starred transform

G1. [PDF pp. 75–79]

The impulse-sampled signal is $x^*(t) = \sum_{k \geq 0} x(kT)\delta(t - kT)$. With $T = 0.5$, the impulse at $t = 1.0$ is the one with $k = 2$, and its *strength* (area) is $x(1.0) = e^{-1} = 0.3679$.

If you missed it. An ideal impulse has zero width and infinite height, so there is no amplitude to read; the coefficient multiplying $\delta(t - kT)$ is an area. This is not pedantry — it is why the ZOH must be modelled separately as $(1 - e^{-Ts})/s$ rather than being absorbed into the sampler.

G2. [PDF pp. 103–106]

The one general identity is

$$[G(s)X^*(s)]^* = G^*(s)X^*(s).$$

A factor that is *already* an impulse train can be pulled out of the star, because sampling an already-sampled signal changes nothing. An unsampled product $G(s)H(s)$ or $G(s)X(s)$ cannot be split: there is no sampler between the factors to break the convolution.

If you missed it. This single rule decides topics I and J almost entirely. If it is not automatic, do nothing else until it is: it is why J1 has $GH(z)$ in the denominator while J2 has $G(z)H(z)$, and why I3 has no pulse transfer function at all.

G3. [PDF pp. 77–78, 103–104]

$X^*(s)$ is periodic along the imaginary axis with $\omega_s = 2\pi/T$. For $T = 0.02$ s, $\omega_s = 2\pi/0.02 = 314.16$ rad/s.

If you missed it. Answering 50 gives $f_s = 1/T$ in hertz. The factor 2π is the whole question, and the same factor reappears wherever ω and f are exchanged.

G4. [PDF pp. 78, 103–104]

Impulse-sampling a unit step gives $X^*(s) = \sum_{k \geq 0} e^{-kTs} = 1/(1 - e^{-Ts})$ as a geometric series. The definition $z = e^{Ts}$ turns e^{-Ts} into z^{-1} , so

$$X(z) = \frac{1}{1 - z^{-1}},$$

which is Table 2–1 entry 3, as it must be.

If you missed it. This is the bridge between the two descriptions: $X^*(s)$ and $X(z)$ are the same object in different variables, which is why a single sampled block may be written either way. It is also why $z = e^{Ts}$ may *not* be substituted into an ordinary continuous plant $G_p(s)$ — the substitution applies to starred quantities, not to unsampled ones.

1.8 H — Data holds and ZOH-plus-plant discretisation

H1–H3. [PDF pp. 111–115]

Equation (3–49) gives the ZOH-plus-plant conversion. With $G_p(s) = 4/(s + 4)$ and $T = 0.5$,

$$\begin{aligned} G(z) &= (1 - z^{-1}) \mathcal{Z} \left\{ \frac{4}{s(s + 4)} \right\} = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{1}{s} - \frac{1}{s + 4} \right\} \\ &= (1 - z^{-1}) \left[\frac{1}{1 - z^{-1}} - \frac{1}{1 - e^{-2}z^{-1}} \right] = 1 - \frac{1 - z^{-1}}{1 - e^{-2}z^{-1}} = \frac{(1 - e^{-2})z^{-1}}{1 - e^{-2}z^{-1}}. \end{aligned}$$

With $e^{-2} = 0.1353$, $G(z) = 0.8647z^{-1}/(1 - 0.1353z^{-1})$.

The DC gain is $G(1) = 0.8647/(1 - 0.1353) = 1$, matching $G_p(0) = 1$.

For a sampled unit step, $Y(z) = G(z)/(1 - z^{-1})$, whose z^{-1} coefficient is 0.8647; so $y[0] = 0$ and $y[1] = 0.8647$.

Check. The ZOH holds the command constant over one period, so $y(T)$ is just the continuous step response of the plant at $t = T$: $1 - e^{-4(0.5)} = 1 - e^{-2} = 0.8647$. The transform and the physics agree, and the agreement costs one line to check.

If you missed it. Losing the z^{-1} is the most common error and the most consequential: a ZOH driving a strictly proper plant *always* produces a one-sample delay, because the output cannot respond before the first hold interval has elapsed. If your $G(z)$ has a nonzero $g[0]$, something is wrong before you go any further.

H4. [PDF pp. 113–114]

Equation (3–49) reads

$$X(s) = \frac{1 - e^{-Ts}}{s} G(s) \implies X(z) = (1 - z^{-1}) \mathcal{Z}\left\{\frac{G(s)}{s}\right\}.$$

Only the numerator factor $1 - e^{-Ts}$ becomes $1 - z^{-1}$ (it is a one-sample delay, and $e^{-Ts} \leftrightarrow z^{-1}$). The $1/s$ belongs to the hold's own integration and must stay *inside* the transform alongside $G(s)$.

If you missed it. The variant $(1 - z^{-1})\mathcal{Z}\{G(s)\}$ drops that $1/s$ and silently discretises the wrong plant. The variant keeping $1 - e^{-Ts}$ outside leaves an s -domain factor multiplying a z -domain expression, which is not a well-formed object at all — a useful sanity filter in itself.

H5. [PDF pp. 113–115]

With $G_p(s) = 1/[s(s+2)]$ and $T = 1$,

$$\frac{G_p(s)}{s} = \frac{1}{s^2(s+2)} = -\frac{0.25}{s} + \frac{0.5}{s^2} + \frac{0.25}{s+2},$$

(coefficients from $1 = As(s+2) + B(s+2) + Cs^2$ at $s = 0$, $s = -2$, and matching s^2). Therefore

$$G(z) = (1 - z^{-1}) \left[\frac{-0.25}{1 - z^{-1}} + \frac{0.5z^{-1}}{(1 - z^{-1})^2} + \frac{0.25}{1 - e^{-2}z^{-1}} \right].$$

Collecting the z^{-1} coefficient: the first term contributes 0, the second 0.5, and the third $0.25(e^{-2} - 1) = -0.2162$. Hence $g[1] = 0.2838$.

Check. Independently, $g[k] = s_r(kT) - s_r((k-1)T)$ where $s_r(t) = -0.25 + 0.5t + 0.25e^{-2t}$ is the plant's continuous step response. Then $s_r(0) = 0$ and $s_r(1) = 0.2838$, so $g[0] = 0$ and $g[1] = 0.2838$. The two routes agree, and $g[0] = 0$ confirms the expected one-sample delay.

1.9 I — Pulse transfer functions, samplers and cascades

I1–I2. [PDF pp. 117–121]

Without an intermediate sampler the blocks must be combined *before* sampling:

$$GH(s) = \frac{1}{(s+1)(s+3)} \implies gh(t) = \frac{1}{2} (e^{-t} - e^{-3t}).$$

Sampling at $T = 0.5$ gives $gh(0) = 0$ and $gh(0.5) = \frac{1}{2}(0.6065 - 0.2231) = 0.1917$.

With an intermediate sampler, each block gets its own pulse transfer function and the sequences convolve: $g[k] = e^{-0.5k}$, $h[k] = e^{-1.5k}$, so the product's coefficients are 1, 0.8297, 0.5530, ...

The two disagree at *every* coefficient, most starkly at $k = 0$, where one is 0 and the other is 1. They are different systems, and no choice of T reconciles them.

If you missed it. The $k = 0$ comparison is the fastest possible proof and worth remembering as a stock argument: $GH(z)$ is strictly proper (two poles, no zeros, so $gh[0] = 0$), while $G(z)H(z)$ is a product of two transforms each starting at 1. If a block-diagram answer ever hinges on which form applies, this settles it in one line.

I3. [PDF pp. 107–110, 126]

The signal reaching the first sampler is $F(s)R(s)$, so the sampled quantity is $[F(s)R(s)]^*$. By G2, a star may be factored only when one factor is already starred, and neither is here. Consequently R cannot be separated from the dynamics and no input-independent ratio $C(z)/R(z)$ exists.

Contrast the repaired configuration: with a sampler placed *before* $F(s)$, $[F(s)X^*(s)]^* = F^*(s)X^*(s)$, and $Y(z) = F(z)X(z)$ does exist.

If you missed it. Sampling the output does not repair a missing input sampler — it only says where the continuous output is observed. Also note what is *not* being claimed: the response to a specified $R(s)$ remains perfectly computable. Only the input-independent ratio fails to exist.

I4. [PDF pp. 95–100]

The plant impulse response satisfies $g(t) = 0$ for $t < 0$, so $g((k-h)T) = 0$ whenever $h > k$ and those terms drop out of the sum. This is causality, and nothing else: an unstable but causal plant has the same summation limits, and a hold changes what g is, not where the sum stops.

1.10 J — Closed-loop configurations, digital controller and PID

J1. [PDF pp. 122–125]

Case 1 of Table 3–1. Derive it rather than recalling it:

$$E(s) = R(s) - H(s)C(s), \quad C(s) = G(s)E^*(s).$$

Substituting, $E(s) = R(s) - H(s)G(s)E^*(s)$. Starring both sides and using $[HGE^*]^* = [HG]^* E^*$:

$$E^* = R^* - [GH]^* E^* \implies E(z) = \frac{R(z)}{1 + GH(z)}, \quad C(z) = \frac{G(z)R(z)}{1 + GH(z)}.$$

There is no sampler between G and H , so the loop term stays welded together as $GH(z) = \mathcal{Z}\{G(s)H(s)\}$. The forward path *is* driven by E^* , which is why the numerator does separate.

J2. [PDF pp. 124–125]

Case 2. Now H is driven by the sampled output, so $E = R - HC^*$ and $[HC^*]^* = H^*C^*$:

$$E^* = R^* - H^*G^*E^* \implies C(z) = \frac{G(z)R(z)}{1 + G(z)H(z)}.$$

If you missed it. J1 and J2 have identical numerators and differ only in whether the denominator reads $GH(z)$ or $G(z)H(z)$. One sampler decides it. Before writing any closed-loop formula, circle every sampler on the diagram and ask, for each product, whether a sampler separates the two blocks; if not, they stay inside one $\mathcal{Z}\{\cdot\}$.

J3. [PDF pp. 125–126]

Case 4 has no sampler after the summing junction, so G_1 is driven by the continuous error. Writing $e = R - HC$ and sampling the signal that actually reaches the sampler, $[G_1e]^*$, gives

$$[G_1e]^* = [G_1R]^* - [G_1G_2H]^* [G_1e]^*, \quad C(z) = \frac{G_2(z)G_1R(z)}{1 + G_1G_2H(z)}.$$

The term $G_1R(z) = \mathcal{Z}\{G_1(s)R(s)\}$ does not factor into $G_1(z)R(z)$, because R passes through G_1 before any sampler. A pulse transfer function is by definition an input-independent ratio, so this system has none — the point the lecture makes explicitly on p. 126.

If you missed it. Note precisely what fails. The system is not unstable, not unanalysable, and not lacking an output: for any *specified* $R(s)$ the response is computable by the same technique. What does not exist is the ratio.

J4, J6. [PDF pp. 134–135]

With $K = 2$, $T = 0.1$, $T_i = 0.5$, $T_d = 0.2$:

$$K_I = \frac{KT}{T_i} = 0.4, \quad K_D = \frac{KT_d}{T} = 4, \quad K_P = K - \frac{KT}{2T_i} = K - \frac{K_I}{2} = 2 - 0.2 = 1.8.$$

For a unit discrete impulse at the error input, read the $k = 0$ coefficient off each term of $G_D(z)$:

$$m[0] = K_P + K_I + K_D = 1.8 + 0.4 + 4 = 6.2, \quad m[1] = K_I - K_D = -3.6.$$

If you missed it. Answering $K_P = 2$ means the $-K_I/2$ correction was missed. It is not cosmetic: the trapezoidal rule averages $e[k-1]$ and $e[k]$, and half of the current sample's contribution is already counted inside the integral term, so the proportional gain must be reduced to compensate. The slide states $K_P < K$ explicitly, which is a free check on your arithmetic. The large $|m[1]|$ is the derivative kick, and it is why K_D is the term that makes PID noise-sensitive.

J5. [PDF pp. 135]

$$G_D(z) = \frac{M(z)}{E(z)} = K_P + \frac{K_I}{1 - z^{-1}} + K_D (1 - z^{-1}).$$

Integration accumulates, giving a pole at $z = 1$; differentiation differences, giving a zero at $z = 1$. Swapping the two — $K_I(1 - z^{-1})$ with $K_D/(1 - z^{-1})$ — inverts both roles at once.

If you missed it. A form with T still visible has double-counted the sampling period: T is already absorbed into $K_I = KT/T_i$ and $K_D = KT_d/T$. Reading the gain definitions and the transfer function as one package prevents this.

After marking

Write your ten topic percentages down before doing anything else, then act on them in the tier order given at the front. Two habits are worth carrying into the quiz regardless of the scores, because they appear in the **Check** lines above more often than any single formula:

1. **Generate a sample.** Any closed form, inverse transform or recurrence solution can be tested at $k = 0$ and $k = 1$ against the original time-domain rule in well under a minute. Every wrong option in B4, E3, F4 and D2 dies to this test.
2. **Check a gain or a limit physically.** A ZOH-plus-plant conversion must reproduce the plant's DC gain; a stable sampled loop must settle where the continuous one would. H2 and H3 are that check applied deliberately, and it catches a lost $(1 - z^{-1})$ factor faster than re-deriving the algebra.

Where a score is low and you are unsure whether the cause is the concept or the arithmetic, re-attempt only the items you missed — the question file keeps the option order fixed, so a second pass measures the same thing.